I. Introduction

The goal of this activity was to create a relational data model using Microsoft Power BI. By organizing multiple tables and establishing relationships, the activity aimed to demonstrate how structured data improves analysis and visualization compared to simple datasets.

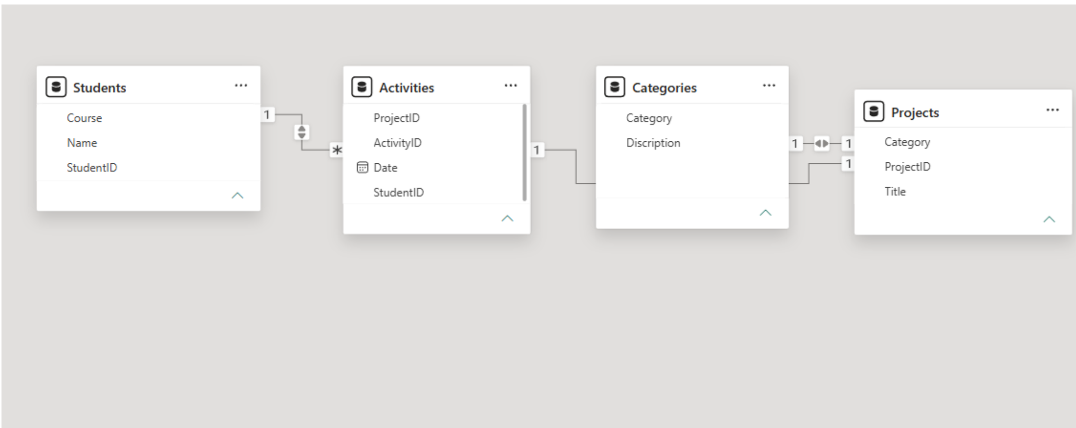
II. Methodology

To complete the activity, the following steps were performed:

- Data Preparation: Created four tables (Students, Activities, Projects, Categories) in Excel with proper identifiers.
- Data Import: Uploaded the dataset into Power BI and verified it using Power Query.
- Model Creation: Accessed Model View and connected tables using common fields such as StudentID, ProjectID, and Category.
- Visualization Setup: Arranged the tables and relationships to form a clear relational schema.

III. Implementation

The activity used Microsoft Power BI to integrate and manage the dataset. Relationships between tables were created to allow interaction among data, and the Model View provided a visual diagram of these connections.



IV. Problem-Solving

Several challenges were encountered during the activity:

- Difficulty locating Model View in the web interface
- Incorrect relationship types (one-to-one instead of one-to-many)
- Improper column headers in one table

Solutions applied:

- Navigated to the correct Model View interface
- Adjusted relationships to one-to-many
- Corrected column headers and matched corresponding fields

V. Conclusion

The activity successfully demonstrated the importance of data modeling in Power BI. Establishing relationships between tables improved data organization and usability. Overall, the activity enhanced understanding of how integrated tools transform raw data into structured and meaningful information.